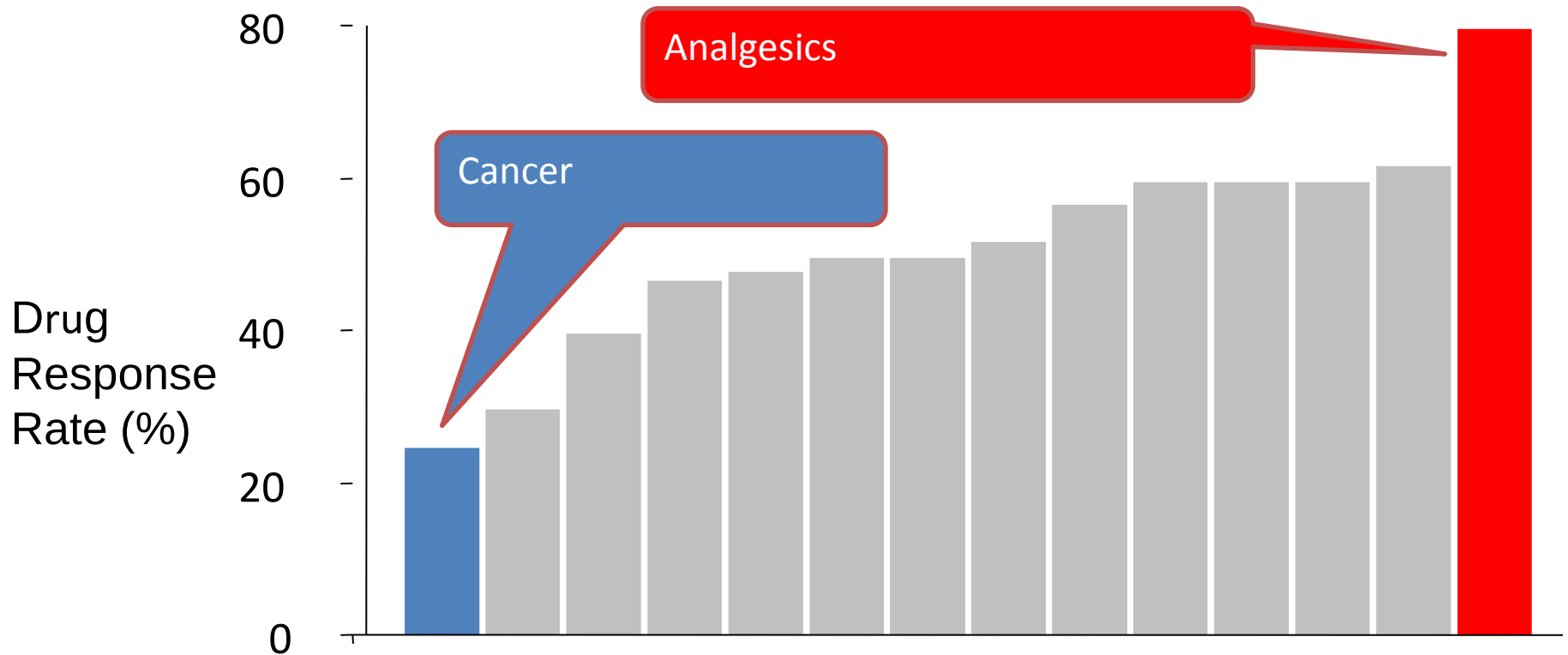


Barcoding cancers: Will this personalise treatment?

Dr James D. Brenton PhD FRCP

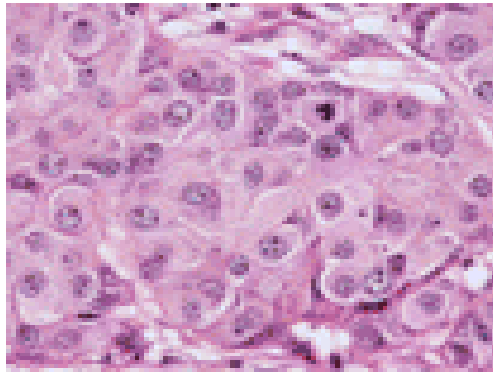
Cancer Research UK Cambridge Institute, University of Cambridge and
Cambridge University Hospitals Foundation Trust and Oncology
Consultant to Pacific Life Re and SCOR Global Life

Not all drugs work for all patients

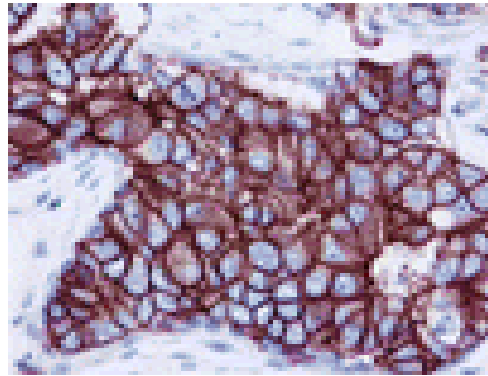


Source: Spears et al., Trends Mol Med, 2001; Lazarou et al., JAMA, 1998,

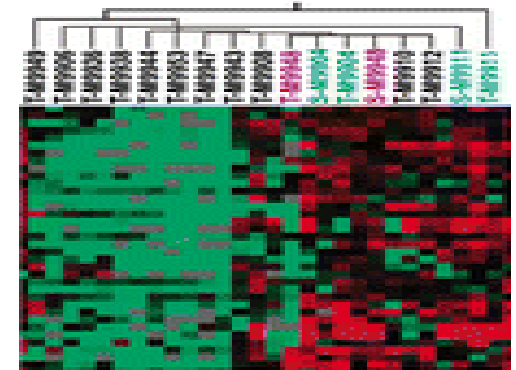
Improving diagnostic markers for personalizing cancer care



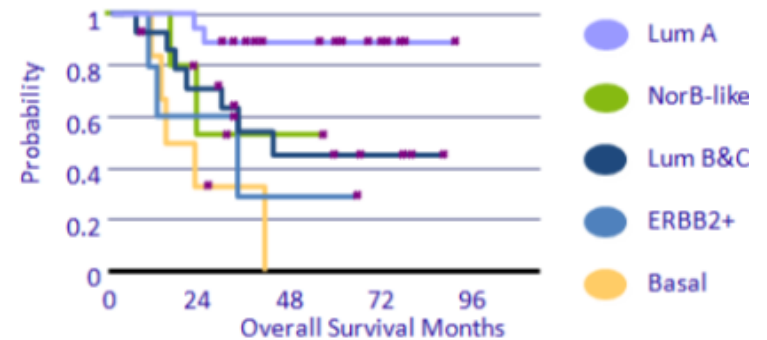
**Morphological
Diagnosis**



**Immunohistochemical
assessment**

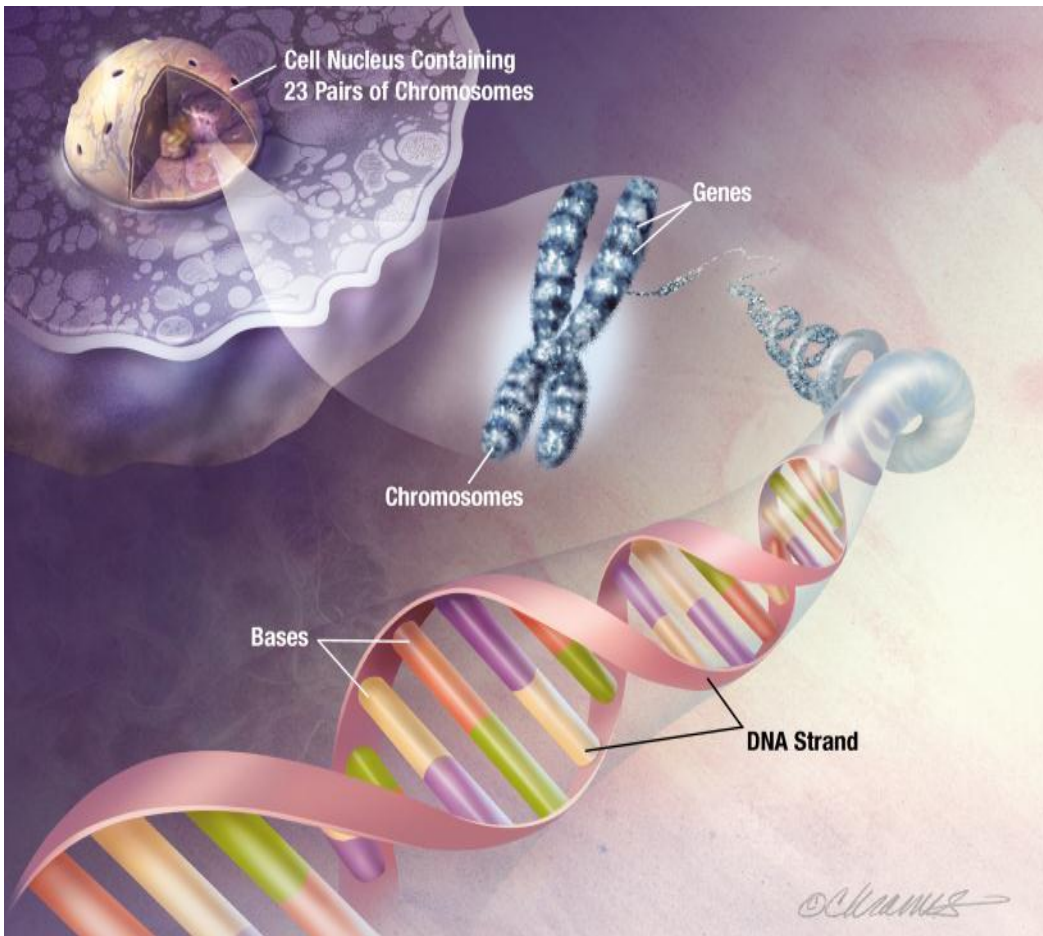


**DNA microarray
analysis**



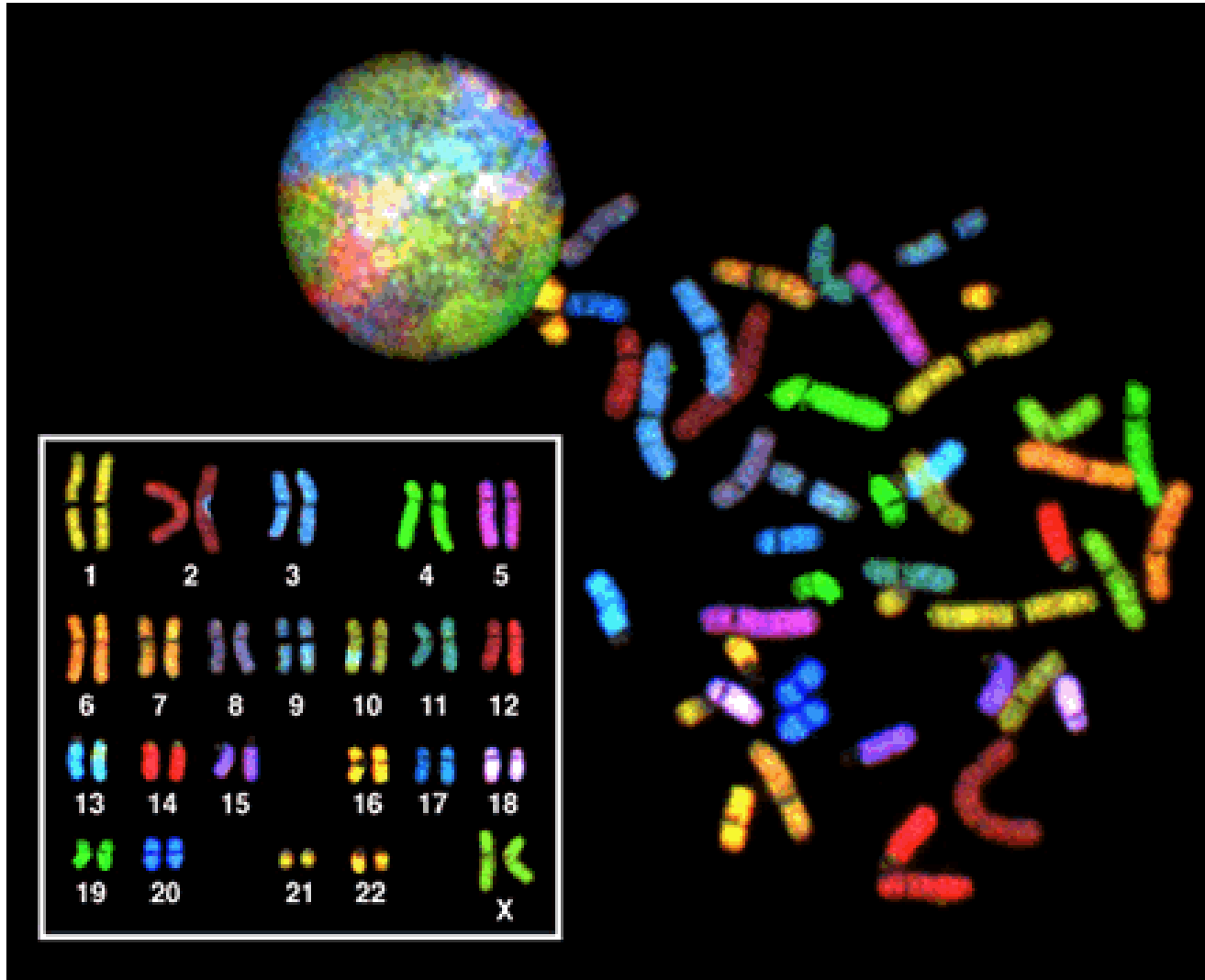
Source: Baselga and Norton, 2002

Cancer is a genetic disease

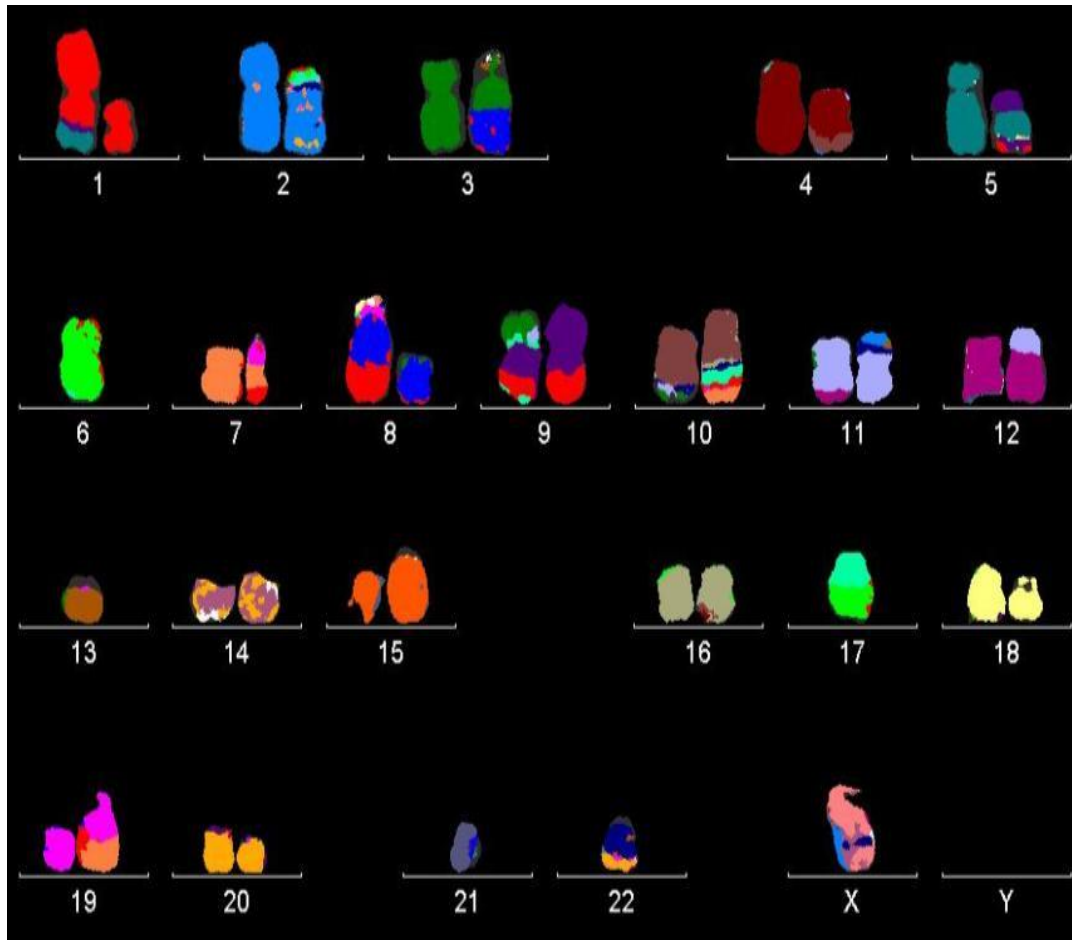


DNA is packed into chromatin and stored as discrete chromosomes within the cell

Normal chromosomes seen using a microscope

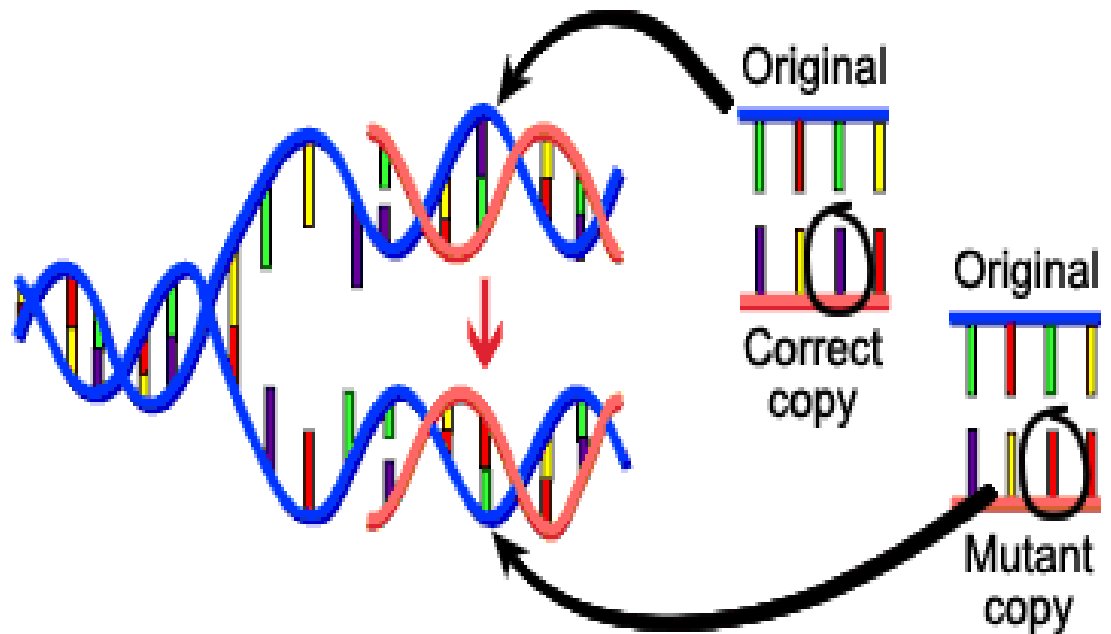
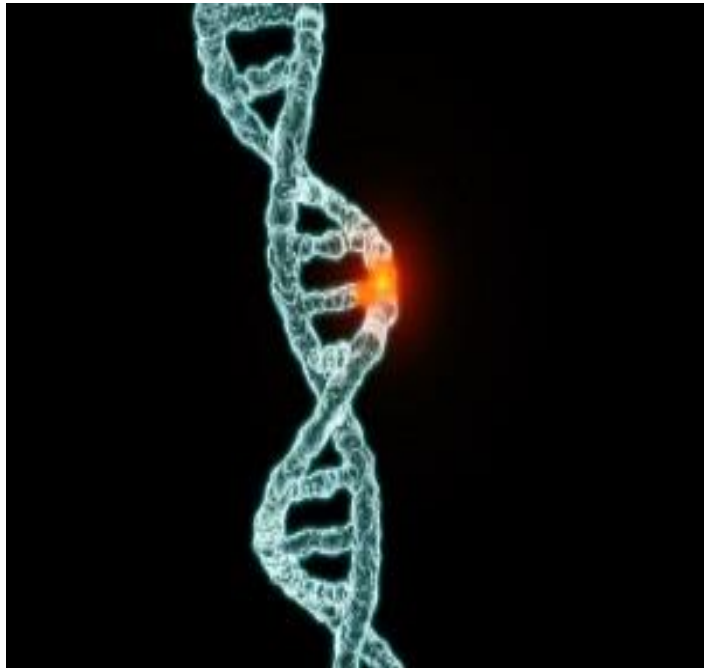


Damage to DNA causes cancer



Fluorescent microscope images of damaged chromosomes from a cancer cell

Sequencing DNA easily shows type of damage caused by cancer



e.g. *BRAF* c.1799T>A (V600E)

New technologies have transformed DNA sequencing

First “next generation” machine

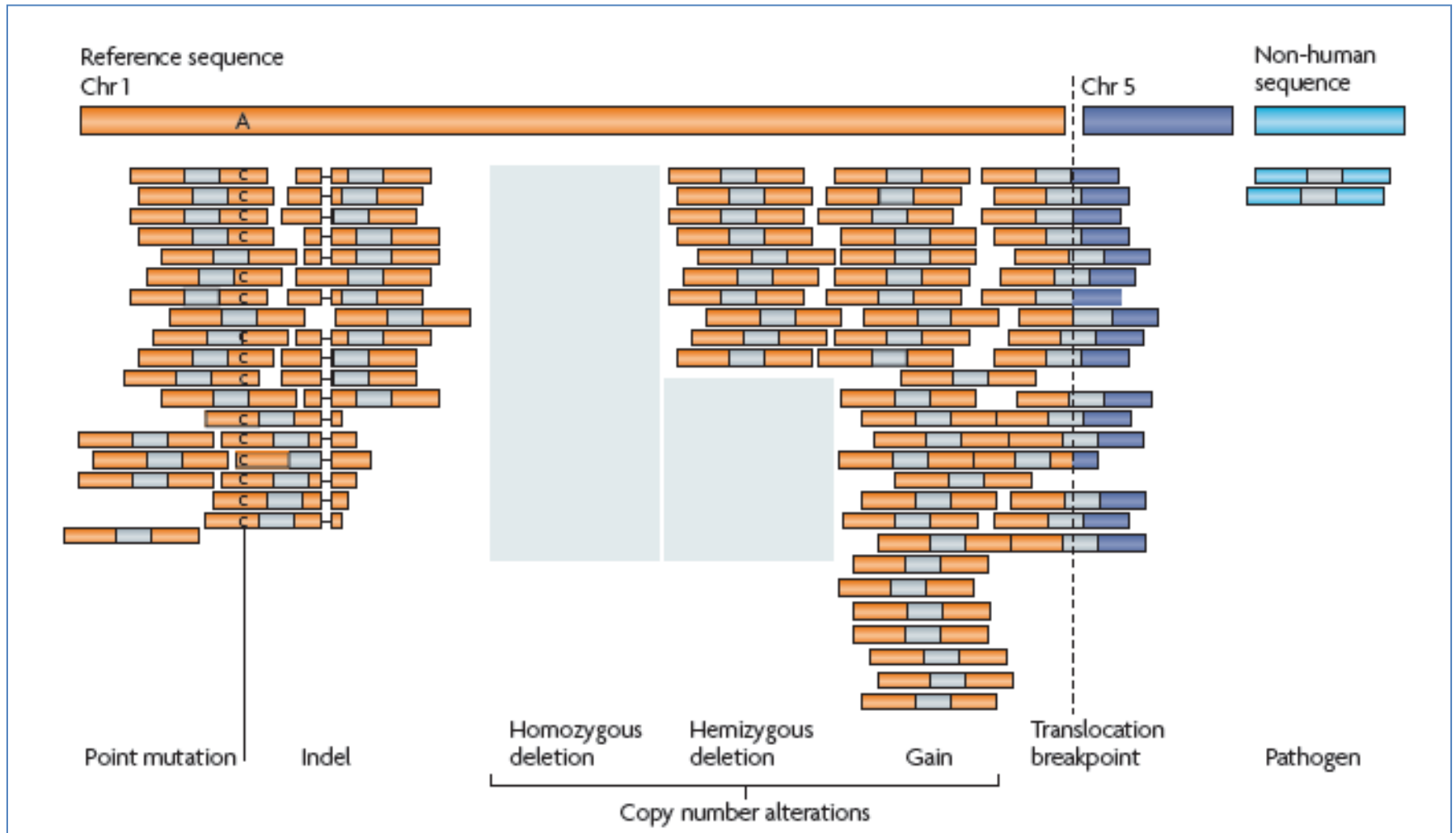


Instrument for diagnostics laboratory



Illumina MiSeq

Next generation cancer sequencing

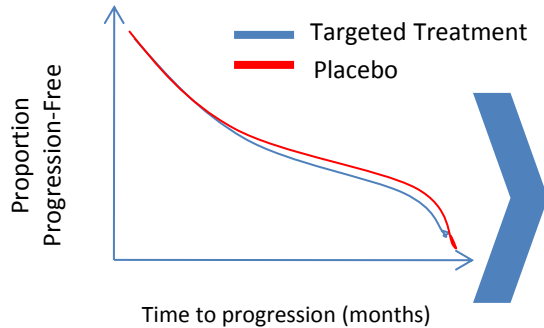


Cancers of the same type are
genetically similar but distinct

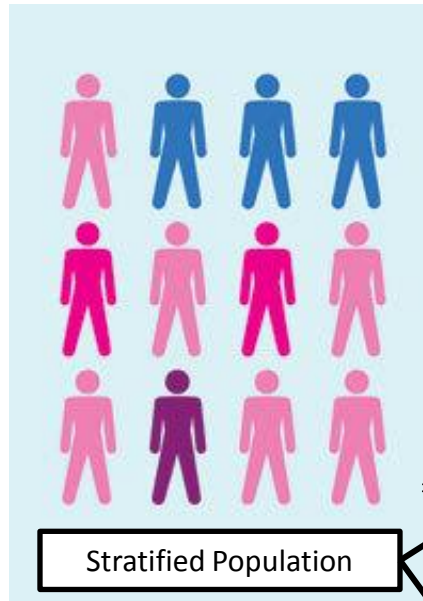


Stratified Medicine

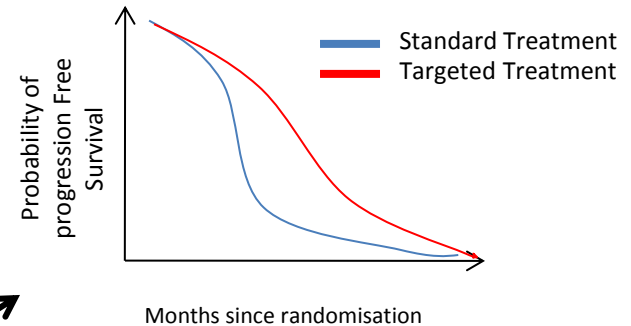
Targeted treatment no more effective than placebo overall



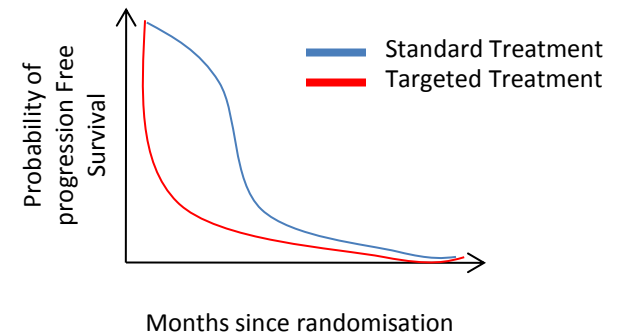
Unstratified Population



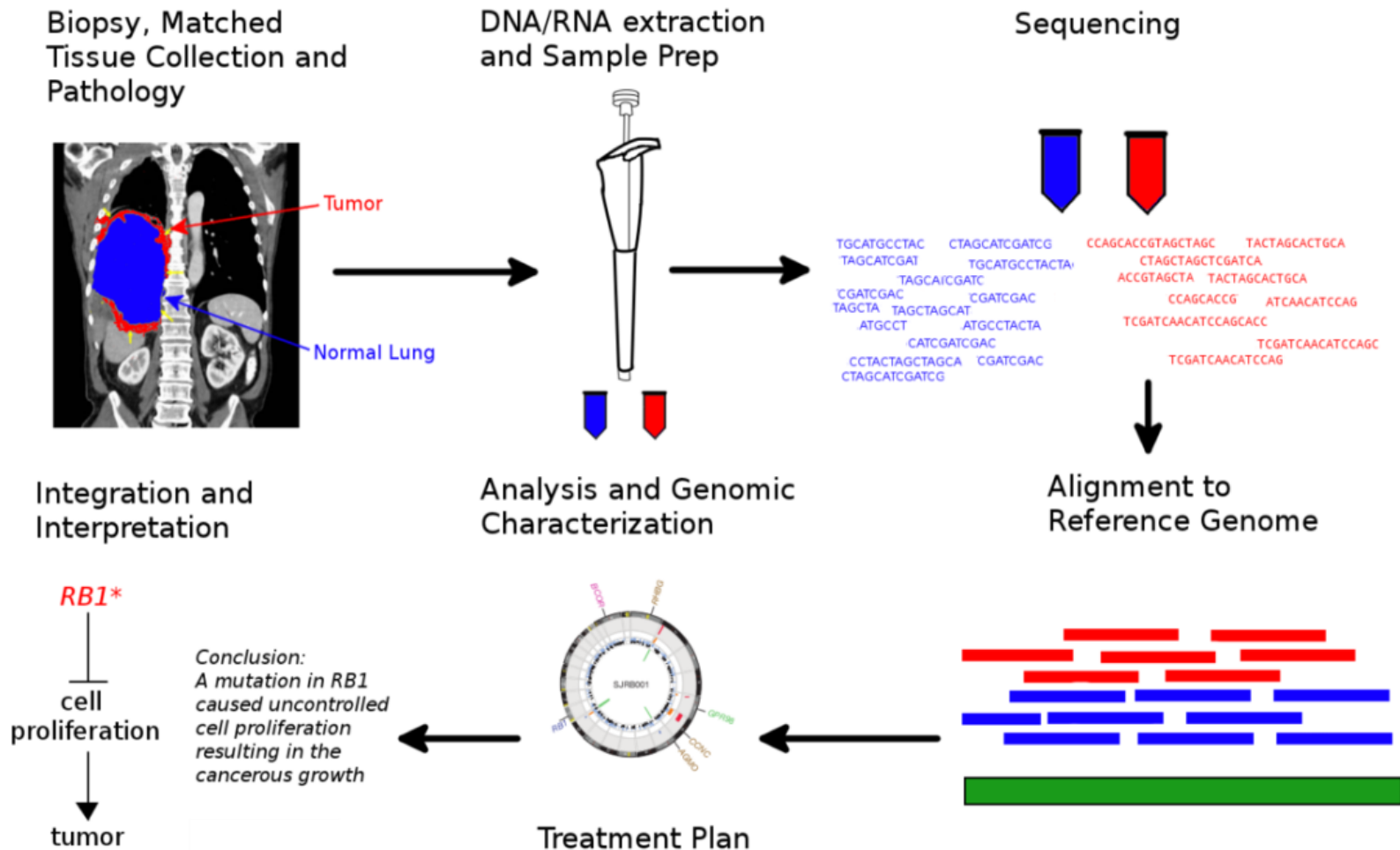
Targeted treatment more effective than standard treatment if mutation is present



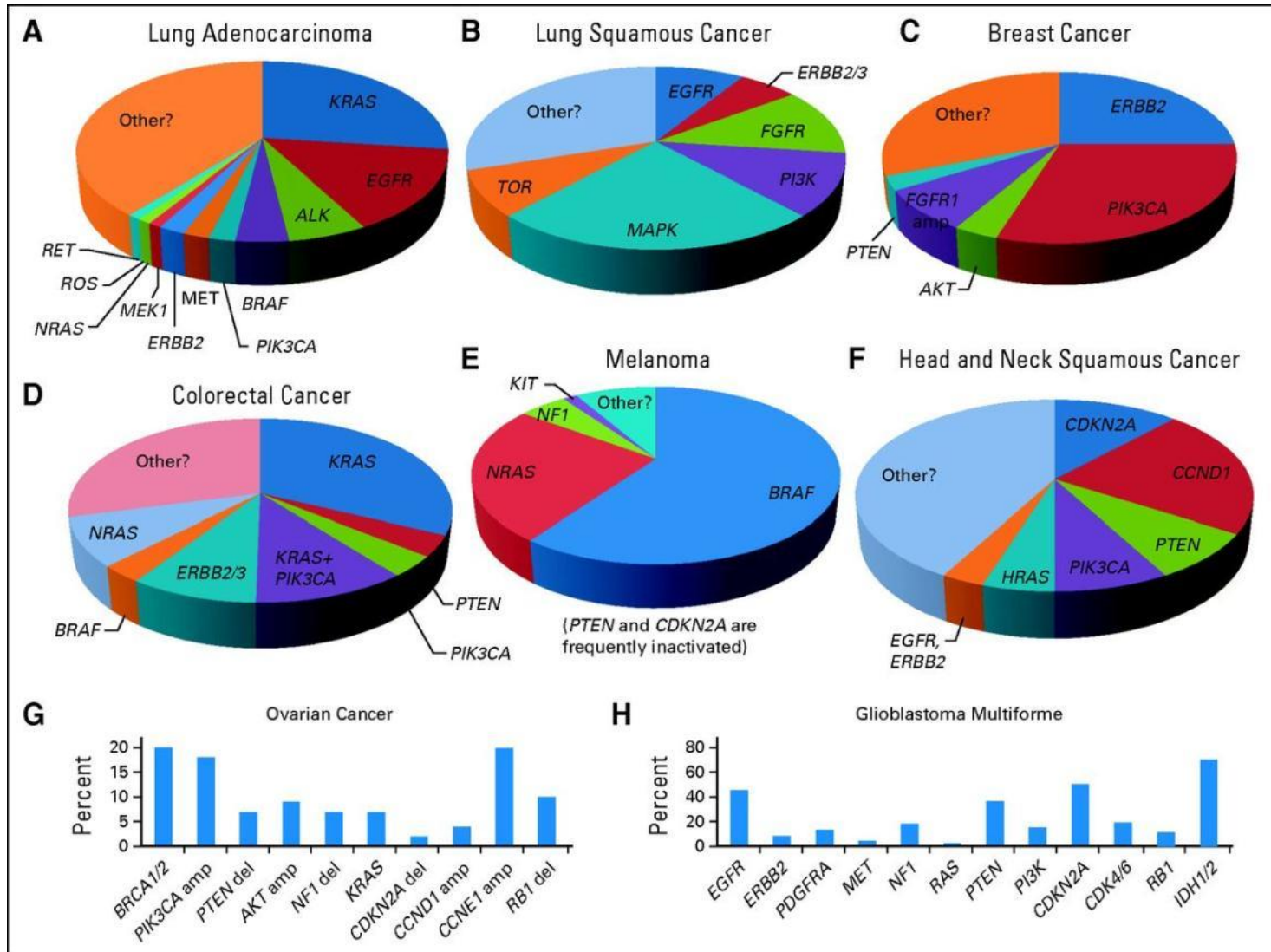
Targeted treatment less effective than standard treatment if mutation is not present



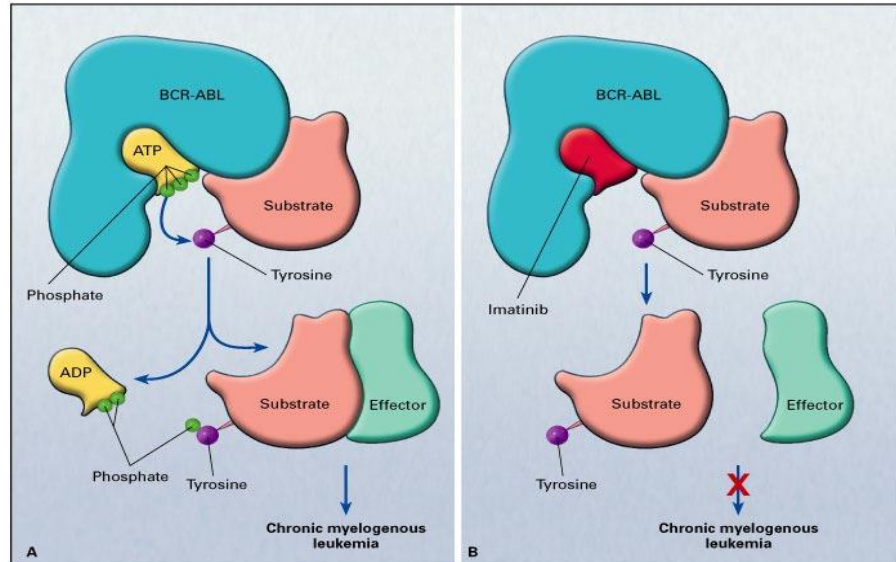
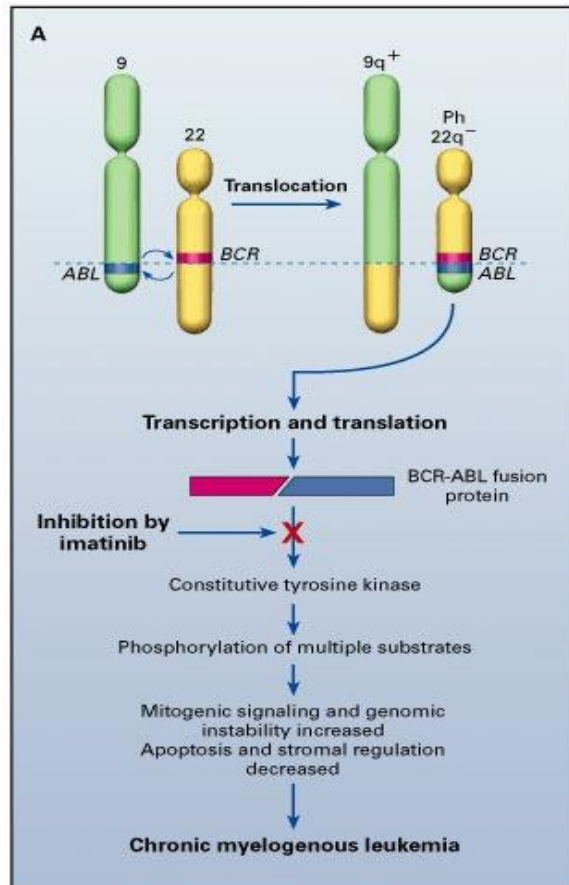
DNA sequencing as a clinical test



Who will potentially benefit?



Chronic Myeloid Leukaemia



The New England Journal of Medicine

Copyright © 2001 by the Massachusetts Medical Society

VOLUME 344

APRIL 5, 2001

NUMBER 14



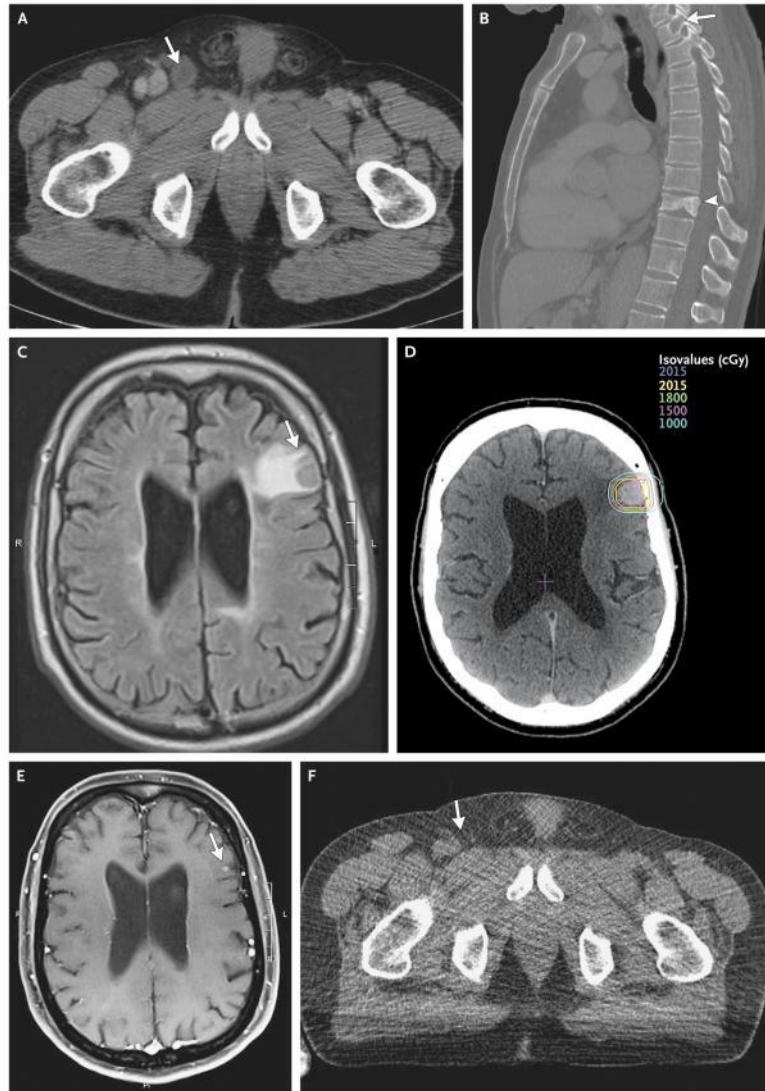
EFFICACY AND SAFETY OF A SPECIFIC INHIBITOR OF THE BCR-ABL TYROSINE KINASE IN CHRONIC MYELOID LEUKEMIA

BRIAN J. DRUKER, M.D., MOSHE TALPAZ, M.D., DEBRA J. RESTA, R.N., BIN PENG, PH.D., ELISABETH BUCHDUNGER, PH.D., JOHN M. FORD, M.D., NICHOLAS B. LYDON, PH.D., HAGOP KANTARJIAN, M.D., RENAUD CAPDEVILLE, M.D., SAYURI OHNO-JONES, B.S., AND CHARLES L. SAWYERS, M.D.

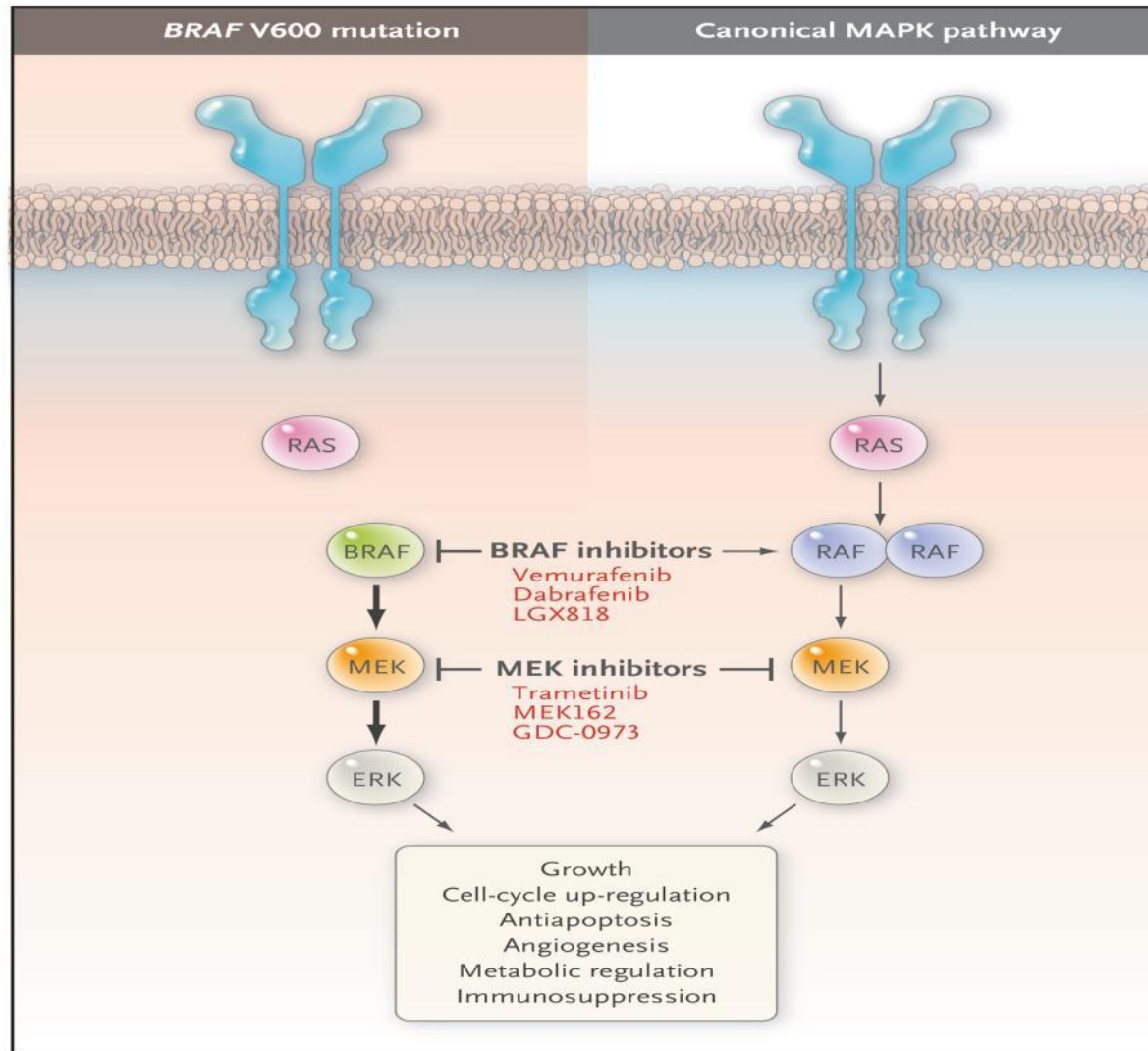
Case presentation

- 68-year-old man
- presented with metastatic melanoma to the groin and pelvis
- 11 years previously had excision of a superficial spreading melanoma on the right leg.

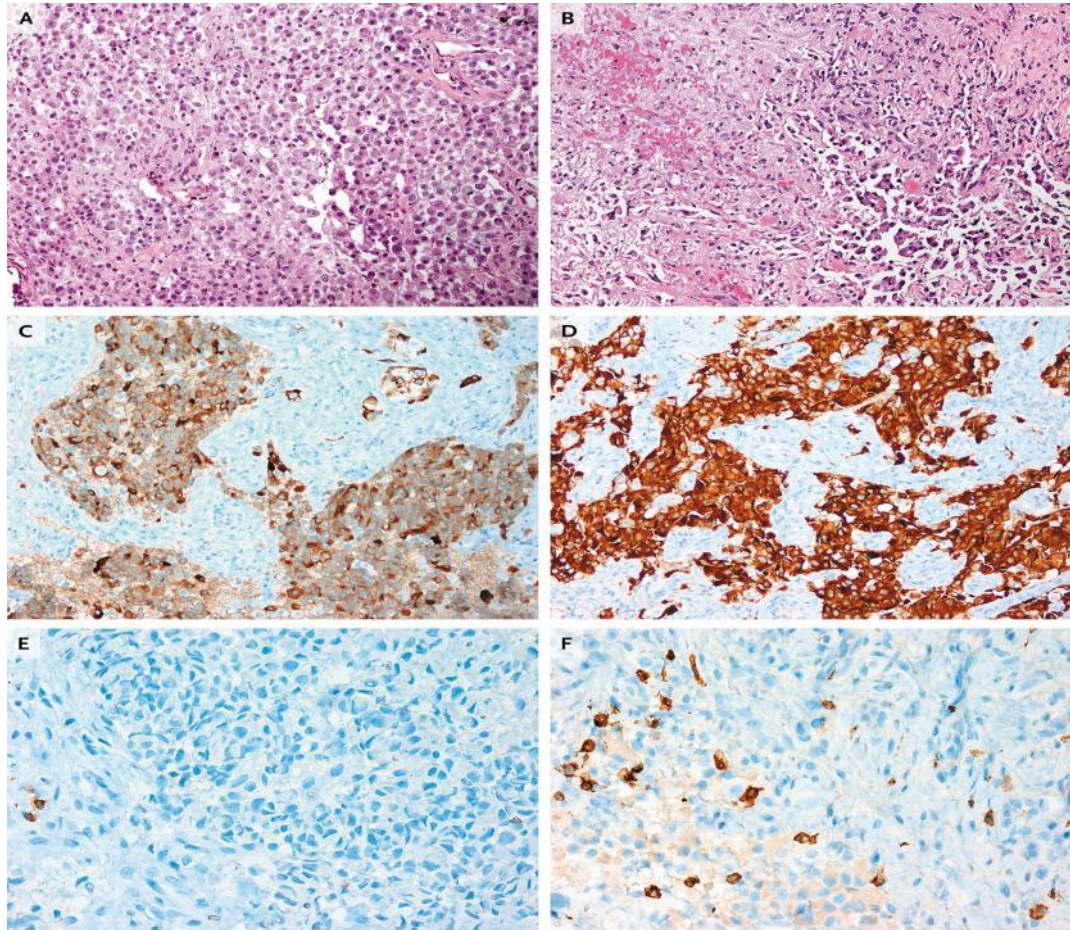
Image Studies



Mitogen-Activated Protein Kinase (MAPK) Pathway



Pathological Analysis of Skin-Biopsy Specimens



Photographs of the Patient after Stereotactic Radiosurgery



Final Diagnosis

- Metastatic melanoma with a BRAF V600E mutation and an response to stereotactic radiosurgery.

Malignant melanoma and secondary mutation

Pre-treatment



15 weeks



23 weeks

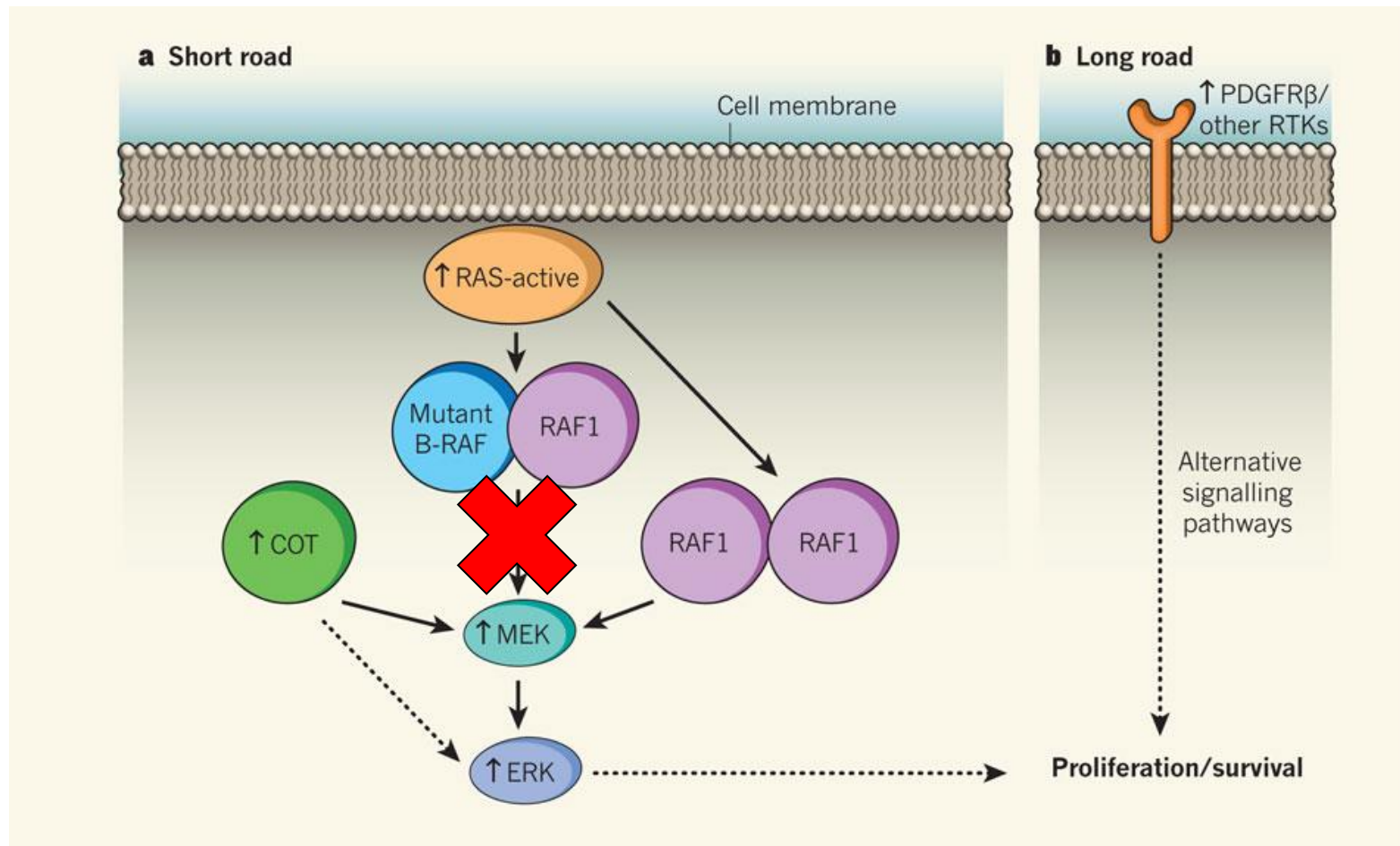


Treatment with vemurafenib

BRAF p.V600E

BRAF p.V600E
MEK1 p.Q57P

Cancer cells become resistant by bypassing “roadblock”



Summary

- DNA sequencing tests are affordable for clinical care and being widely implemented
- Sequencing will provide rapid changes in classification for all cancers
- Currently, only a small proportion of patients will see marked survival benefit
- Survival estimates (and pricing/claims) need to be in step with emerging data from personalized medicine

Thank you

- Dr James D. Brenton PhD FRCP

Cancer Research UK Cambridge Institute, University of
Cambridge and Cambridge University Hospitals Foundation
Trust and Oncology Consultant to Pacific Life Re and SCOR
Global Life